Attainable concordance signature on Swiss Market Index

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Abstract

The Swiss Market Index contains 20 stocks with only 18 stocks' data completed, and we are interested in the attainable concordance signature on the set of these 20 stocks. We preprocess the data into an elliptical distribution, reduce the demension of the dataset to simplify the process, and study the set of attainable concordance signature on some subsets $\mathcal{D}$ of the data. The results show that such attainable sets form a convex hull, and that the attainable concordance signature on a particular subset I of $\mathcal{D}$ would not change as new stocks are added into $\mathcal{D}$. Furthermore, in the case of our data, the common assumption of higher-order independence would not be attainable.

1 Introduction

In this project, I studied the stock prices of the Swiss Market Index (SMI) and its components. SMI is Switzerland's blue-chip stock market index made up of the 20 largest and most liquid Swiss Performance Index stocks. I gained exposure to this index in book McNeil et al. (2015), and did some exploration to gain a better understanding of copula and rank correlation. I downloaded the adjusted prices of the 20 component stocks of SMI in year 2020 from Yahoo Finance and used them as the dataset for this project. I aim to study the dependence among these 20 stocks since they could be treated as the marginal distributions of SMI.

The raw data is difficult to use. Some stocks have only one or two missing values, while some stock data are partially missing or fully missing because they were newly added or represented in a different currency unit. I applied

some preprocessing technology introduced in book McNeil et al. (2015), such as taking log returns, fitting GARCH models and using residuals.

The measure of association that I focus on in this project is probabilities of concordance, κ_I, a higher-order measure of association that is defined similar to pairwise Kendall's tau correlation. It allows us to measure the association among a subset I with cardinality larger than 2. The definition could be found in McNeil et al. (2020), as well as the definition of concordance signature, which is the vector of probabilities of concordance $(\kappa_I : I \in \mathcal{E}(\mathcal{D}))$. The pairwise probabilities of concordance κ_I is highly related to Kendall's tau correlation τ: $2\kappa_{\{i,j\}} - 1 = \tau_{\{i,j\}}$. And for elliptically distributions, the concordance signature of a set $\mathcal{D}$ could be fully determined by its Kendall's tau matrix $\mathcal{P}$.

However, due to the missing values, I am unable to compute the concordance signature or Kendall's tau matrix directly from the data, so I could not obtain the accurate concordance signatures. Insteed, I could focus on the sets of attainable concordance signature on the dataset. Such sets would form a convex hull and the methodology used to obtain it is introduced in paper McNeil et al. (2020).

Another problem during my study is the high dimensionality of the dataset. There are 20 stocks in total, with 18 stocks with almost no missing values, one stock with its data half missing, and one stock with its entire data missing. The running time would increase exponentially with dimension, so, by the algorithm introduced in paper Perreault et al. (2019), I partitioned these 18 stocks with their data completed into 13 clusters. This partition satisfies the Partial Exchangeability Assumption (PEA), and thus the components in the same cluster could be considered to have the same Kendall's tau correlation. Because of this, we could choose one stock from each cluster to represent the rest of the stocks, and thus reduce the dimension.

In this report, I would first describe my data, introduce the methodology and preprocess it into an elliptical distribution without auto-correlations. Then I would reduce the dimension of the dataset in section 3. In section 4, I would focus on the set of attainable concordance signature on some subsets of the dataset. I would compute the the convex hull of the attainale concordance signature on some sets $\mathcal{D}$, study the impact of adding more stocks into $\mathcal{D}$, and test the assumption of higher order independence on $\mathcal{D}$.

2 Data preprocessing

2.1 Data description

The dataset I used in this project is the adjusted prices of the 20 component stocks of SMI in year 2020. Stock names and tickers are listed in table 1, sorted by their weightings in the index. Note that during this project, I refer to each stock by its ticker.

Table 1: Components of Swiss Market Index

Rank	**Name**	**Industry**	**Ticker**
1	Nestlé	Food	NESN
2	Novartis	Pharmacy	NOVN
3	Hoffmann-La Roche	Pharmacy	ROG
4	Zurich Inserance Group	Insurance	ZURN
5	ABB	Electrical equipment	ABBN
6	UBS	Banks	UBSG
7	Lonza	Chemistry	LONN
8	Givaudan	Chemistry	GIVN
9	Richemont	Luxury Goods	CFR
10	Sika	Chemistry	SIKA
11	LafargeHolcim	Building Materials	LHN
12	Swisscom	Telecommunications	SCMN
13	Alcon	Pharmacy	ALC
14	Credit Suisse	Banks	CSGN
15	Partners Group	Private equity	PGHN
16	Swiss Re	Insurance	SREN
17	Geberit	Sanitary engineering	GEBN
18	SGS	Services	SGSN
19	Swiss Life Holding	Insurance	SLHN
20	Swatch Group	Watches	UHR

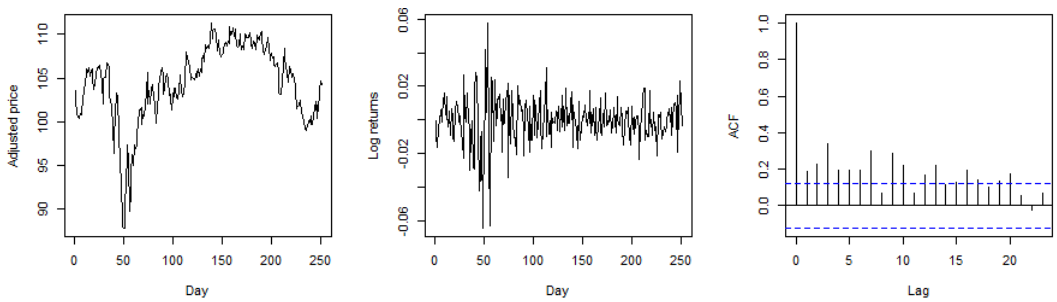
Data for stock PGHN before September 18th is not included, because it was newly added to this index by this date and the data before that date could not be considered as a marginal distribution of SMI. Also, data for stock ALC is treated as fully missing, because its price is represented in currency USD, while the prices of all other stocks are represented in currency CHE. The other 18 stocks only have one or two missing instances, so, instead of treating them as missing, I would estimate them by averaging the adjusted prices of the day

before and the day after. Note that prices on weekends and holidays are not included, because stocks only trade on business days. After cleaning, data for ALC is entirely missing, PGHN has 70 data instances, and the other 18 stocks each have 251 data instances.

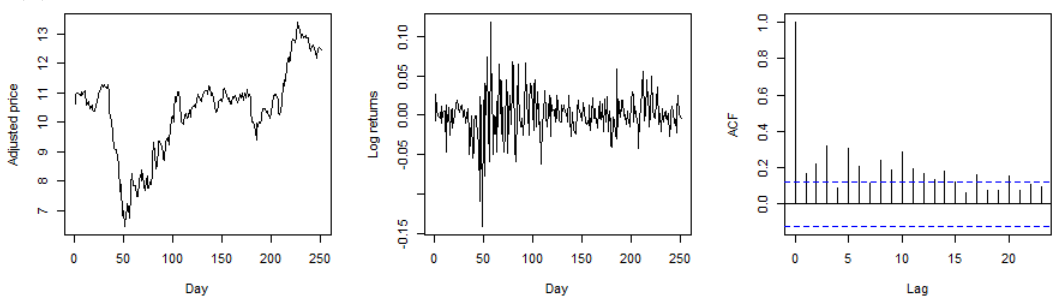
An example of raw data of NESN and UBSG is plotted in figure 1. One could not observe any obvious trend or pattern from these two time series plots, nor discover any suitable models that can describe them. So, I took log returns

$$\text{log returns} = \log(\frac{\text{price}_{\text{now}}}{\text{price}_{\text{before}}}),$$

then plotted them next to the raw data. For financial data, it is common that its raw log returns seem to be without auto-correlation, so I plotted the auto-correlograms of the asbolute values of log returns in figure 1 on the right. Auto-correlations indeed exist.



(a) Raw data, log returns, and auto-correlogram of absolute log returns of NESN



(b) Raw data, log returns, and auto-correlogram of absolute log returns of UBSG

Figure 1: Some explorations on the dataset

For the purpose of futher convenience, I would prefer to deal with data following an elliptical distribution and without auto-correlation. So, I would fit some appropriate models to each stock and then use the residuals.

From the log returns in figure 1, one could discover a continuous variation without a definite trend or pattern, with some spikes. This type of data could

usually be captured by a GARCH model. So in section 2.2, I would talk about the methodology used for fitting and checking a GARCH(1,1) model, introduced in book (McNeil et al., 2015, 4.3).

2.2 Garch(1,1) Models

Lower-order GARCH models such as GARCH(1,1) are most widely used when estimating financial data. A GARCH(1,1) model is defined as:

Definition 1 *Let $(Z_t)_{t\in\mathbb{Z}}$ be $SWN(0,1)$ (strict white noise with mean 0 and variance 1), process $(X_t)_{t\in\mathbb{Z}}$ is a GARCH(1,1) if it is strictly stationary and if for all $t \in \mathbb{Z}$ and some strictly positive-valued process $(\sigma_t)_{t\in\mathbb{Z}}$, it satisfies the following equations*

$$\begin{aligned} X_t &= \sigma_t Z_t, \\ \sigma_t^2 &= \alpha_0 + (\alpha_1 Z_{t-1}^2 + \beta)\sigma_{t-1}^2, \end{aligned} \tag{1}$$

where $\alpha_0 > 0$, $\alpha_1 \geq 0$, $\beta \geq 0$.

This could be solved by

$$X_t = Z_t \sqrt{\alpha_0 \left(1 + \sum_{i=1}^{\infty} \prod_{j=1}^{i} (\alpha_1 Z_{t-j}^2 + \beta)\right)}.$$

In practice, the most widely used approach to fit a GARCH model is maxium likelihood. Suppose we have a total of $n+1$ data values $X_0, ..., X_n$, then the joint density of $X_1, ...X_n$ conditioning on realized values of X_0 and σ_0 is

$$f_{X_1,...X_n|X_0,\sigma_0}(x_1, ...x_n|x_0, \sigma_0) = \prod_{t=1}^{n} f_{X_t|X_{t-1},...,X_0,\sigma_0}(x_t|x_{t-1}, ..., \sigma_0).$$

This gives us a conditional likelihood

$$L(\alpha_0, \alpha_1, \beta; X) = \prod_{t=1}^{n} \frac{1}{\sigma_t} g(\frac{X_t}{\sigma_t}), \quad \sigma_t = \sqrt{\alpha_0 + \alpha_1 X_{t-1}^2 + \beta + \sigma_{t-1}^2}.$$

Since σ_0 is not actually observed, we usually choose a starting value, such as the sample variance of $X_1, ..., X_n$.

It is usual to check fitted GARCH models using standardized residuals. First, we should check if the residuals behave like an SWN by constructing correlograms and applying portmanteau tests, as described in detail in (McNeil et al., 2015, 4.2). If SWN hypothesis is not rejected, then we would investigate the validity of the distribution (normal or t) used in maxium likelihood fitting, using QQplots.

2.3 Fitting GARCH(1,1) models to data

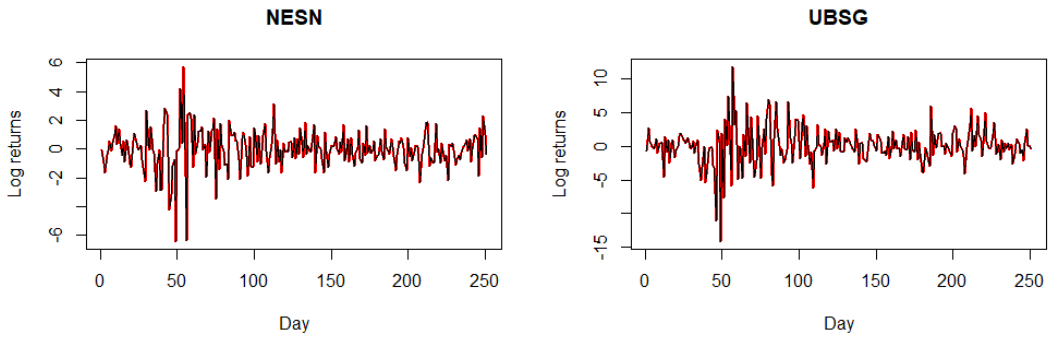
Excluding ALC, to the rest of the 19 stocks, I each fitted a standard GARCH(1,1) model with scaled t or normal innovations as decribed in equation 1, using R package 'rugarch'. Below is an example of the code and outputs of fitting and checking models to stocks NESN and UBSG.

```
library(rugarch)
# model1 is standard GRACH(1,1) using scaled t innovation
model1 = ugarchspec(variance.model=list(model="sGARCH",
    garchOrder=c(1,1)),
        mean.model = list(armaOrder=c(0,0),include.mean = FALSE),
            distribution.model = "std")
# model2 is standard GARCH(1,1) using normal innovation
model2 = ugarchspec(variance.model=list(model="sGARCH",
    garchOrder=c(1,1)),
        mean.model = list(armaOrder=c(0,0),include.mean = FALSE),
            distribution.model = "norm")

gfit_NESN = ugarchfit(spec = model2, data = smi$NESN$data)
fit_NESN = gfit_NESN@fit$sigma*gfit_NESN@fit$z #fitted value
gfit_UBSG = ugarchfit(spec = model1, data = smi$UBSG$data)
fit_UBSG = gfit_UBSG@fit$sigma*gfit_UBSG@fit$z #fitted value
```

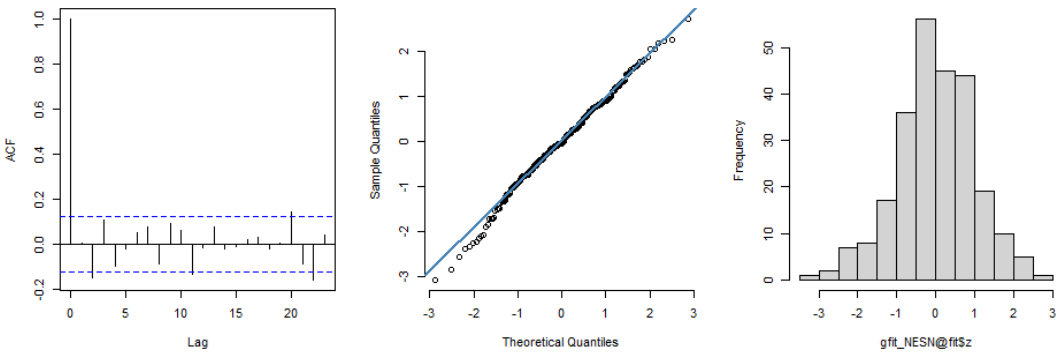
The parameters are given by the following, where omega represents α_0 in equation 1, shape represents the parameter of scaled t innovation.

```
> gfit_NESN@fit$coef

     omega     alpha1      beta1
0.06975943 0.15296103 0.80669880

> gfit_UBSG@fit$coef

    omega    alpha1     beta1     shape
0.1917016 0.1473971 0.8428919 4.3593145
```

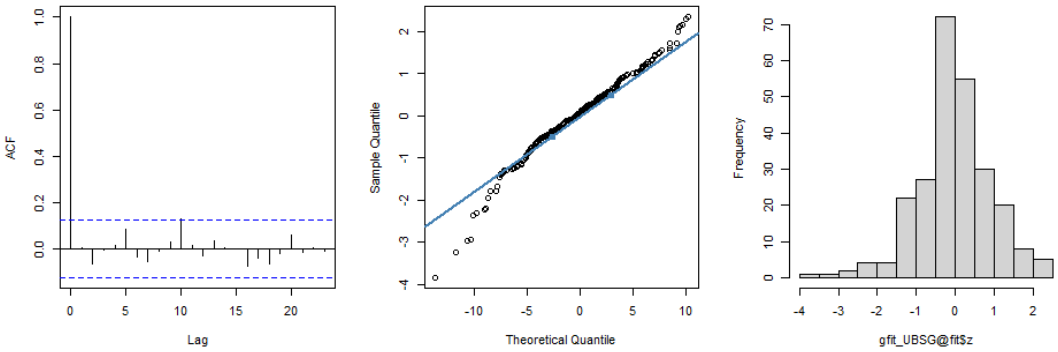
The fitted values are shown in red curves over the original log returns in figure 2a, one could say that the models indeed capture the data well. Then, a check for SWN properties of residuals is applied. The auto-correlograms of absolute residuals are checked. For both stocks, auto-correlations are near 0 for any lag other than zero, so randomness is insured. Portmanteau tests



(a) Fitted log returns for NESN and UBSG



(b) Auto-correlogram for absolute residuals, QQplots and histogram for residuals of NESN.



(c) Auto-correlogram for absolute residuals, QQplots and histogram for residuals of UBSG.

Figure 2: Checking the goodness of fit and residuals

of strict white noise are also applied; the p-values are 0.3022 and 0.06628 correspondingly, so null hypothesis of strict white noise is not rejected. The validity of distributions used in the maxium likelihood fitting, which is normal for NESN and scaled t for UBSG, is checked in figure 2b and 2c using QQplots

and histograms.

```
> Box.test(gfit_NESN@fit$z, lag = 1, type = "Ljung-Box", fitdf = 0)

p-value = 0.4939

> Box.test(gfit_UBSG@fit$z, lag = 1, type = "Ljung-Box", fitdf = 0)

p-value = 0.06628
```

3 Reducing dimension

3.1 A quick exploration on Kendall's correlation

Excluding ALC and PGHN, the Kendall's tau matrix of the 18 stocks is estimated using the dataset and is visualized in figure 3a, with a higher value represented by a darker color. The highest Kendall's correlation 0.731 is attained by UBSG and CSGN in bank industry, while the lowest correlation 0.101 is attained by UBSG and LONN.

The length of the concordance signature on a set $\mathcal{D}$ with dimension d is 2^{d-1}. So a high dimensionality would exponentially increase the running time and complicate the interpretation of results. So in this section, I would use the algorithm introduced in Perreault et al. (2019) to reduce the dimension of kendall's tau matrix for these 18 stocks. Details for the methodology in section 3.2 could be found in Perreault et al. (2019).

3.2 Block structure

We first introduce a very important property of a partition $\mathcal{G}$ of a dataset, the Partial Exchangeability Assumption (PEA), and its two properties.

Definition 2 *For vectors $(X_1, ..., X_d)$ with continuous margins $F_1, ..., F_d$ and copula C, A partition $\mathcal{G} = \{\mathcal{G}_1, ..., \mathcal{G}_K\}$ of $\{1, ..., d\}$ satisfies the* ***Partial Exchangeability Assumption*** *(PEA) if for any cluster preserving permutation π of $1, ..., d$ and all $u_1, ..., u_d \in [0, 1]$, the following equation holds:*

$$C(u_1, ..., u_d) = C(u_{\pi(1)}, ..., u_{\pi(d)}).$$

Cluster preserving permutation π is such that $j \in \mathcal{G}_k$ iff $\pi(j) \in \mathcal{G}_k$ for all $j \in \{1, ..., d\}$ and $k \in \{1, ..., K\}$.

Proposition 1 *A partition $\mathcal{G}$ that fulfills the PEA has two properties:*

1. *Any refinement of $\mathcal{G}$ satisfies the PEA too.*

2. $$(\hat{\boldsymbol{\tau}} - \tilde{\boldsymbol{\tau}})^\top (\hat{\boldsymbol{\Sigma}}^{-1})(\hat{\boldsymbol{\tau}} - \tilde{\boldsymbol{\tau}}) \rightsquigarrow \chi^2_{p-L},$$

   where $\hat{\boldsymbol{\Sigma}}$ is an estimator of the finite sample variance $\boldsymbol{\Sigma}$ and is invertible; $\hat{\tau}$ is the nonparametric estimator of τ; $\tilde{\tau} = \boldsymbol{\Gamma}\hat{\tau}$ is defined in Perreault et al. (2019).

Using these two properties, Perreault et al. (2019) gives an algorithm that finds the best partition $\mathcal{G} = \{\mathcal{G}_1, ..., \mathcal{G}_K\}$ for each $K = \{1, ..., |\mathcal{G}|\}$. Recursively, it finds the best refinement of the partition found in the previous iteration. The algorithm is listed below.

Algorithm 1 Construction of the path $\mathcal{P}$

Fix $w \in [0, 1]$ and set $\mathcal{G}^{(d)} = \{\{1\}, ..., \{d\}\}$, $\tilde{\boldsymbol{\Sigma}}_w^{(d)} = \tilde{\boldsymbol{\Sigma}}(\hat{\boldsymbol{\Sigma}}|\hat{\boldsymbol{\tau}}, \mathcal{G}^{(d)}, w)$.
for $i \in \{d-1, ..., 1\}$ **do**
    Set $\mathcal{G}^{(i)} = \underset{\mathcal{G}^*:\tau_{\mathcal{G}^*} \subset \tau_{\mathcal{G}^{(i+1)}}, |\mathcal{G}^*|=i}{\arg\min} \mathcal{L}\big(\tilde{\boldsymbol{\tau}}(\hat{\boldsymbol{\tau}}|\mathcal{G}^*)|\hat{\boldsymbol{\tau}}, \tilde{\boldsymbol{\Sigma}}_w^{(i+1)}\big)$. ▷ Structure selection
    Set $\tilde{\boldsymbol{\Sigma}}_w^{(i)} = \tilde{\boldsymbol{\Sigma}}(\hat{\boldsymbol{\Sigma}}|\hat{\boldsymbol{\tau}}, \mathcal{G}^{(i)}, w)$.
Return $\mathcal{P} = \{\mathcal{G}^{(d)}, ..., \mathcal{G}^{(1)}\}$.

Notation $\tilde{\boldsymbol{\Sigma}}$ is defined in Perreault et al. (2019). $\mathcal{L}(\mathbf{t}|\hat{\boldsymbol{\tau}}, \boldsymbol{\Sigma}) = (\hat{\boldsymbol{\tau}} - \boldsymbol{t})^\top \boldsymbol{\Sigma}^{-1}(\hat{\boldsymbol{\tau}} - \boldsymbol{t})$ is the loss fuction for the orachle estimator. And the p-value for hypothesis that $\mathcal{G}^{(i)}$ satisfies PEA is given by $\alpha_{(i)} = \mathbb{P}[\chi^2_{p-L_i} > \mathcal{L}(\tilde{\boldsymbol{\tau}}^{(\boldsymbol{i})}|\hat{\boldsymbol{\tau}}, \tilde{\boldsymbol{\Sigma}}^{(\boldsymbol{i})})]$.

For a particular partition $\mathcal{G}$, for any block $\mathcal{B}_l$ of the orginal Kendall's tau, the smoothed Kendall's tau of each block is given by $\tilde{\boldsymbol{\tau}}(\hat{\boldsymbol{\tau}}|\mathcal{G})_r = \sum_{s \in \mathcal{B}_l} \hat{\boldsymbol{\tau}}_s / |\mathcal{B}_l|$ (Perreault et al., 2019, Theorem 1).

3.3 Reducing dimension

Using Algorithm 1, I smoothed the Kendall's tau matrix into dimension 13 and visualized it in figure 3c, choosing the least number of clusters that would not reject the hypothesis of PEA. The code is attached below, and details could be found in https://github.com/samperochkin/learning-block-exchangeability.

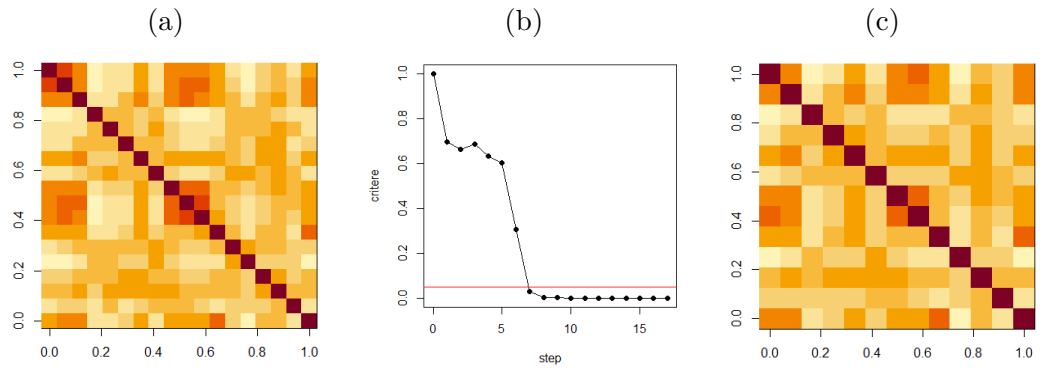


Figure 3: (a) represents the Kendall's tau matrix of the 18 stocks. (b) plots the p-value of hypothesis PEA at each iteration. (c) represents the smoothed Kendall's tau matrix.

```
X = read.table('residuals.csv')
Tau.hat <- cor.fk(X) #original Kendall's tau matrix
# Run the algorithm
path <- pathBuilder(Tau.hat, Theta.hat, nrow(X), 1)
#Plot p-value against iteration
a <- stoppingCriterion(Tau.hat, path$D, path$S)
plot(0:(ncol(X)-1),a,pch=19,xlab="step",ylab="critere")
lines(0:(ncol(X)-1),a)
abline(h=.05,col="red")

#Choose clusters and generate smoothed tau
D.final = path$D[[6]]
new.list <- sapply(1:ncol(D.final), function(i){
  which(D.final[,i] == 1)
})
pairs <- t(combn(1:length(new.list),2))
tt.small <- sapply(1:nrow(pairs), function(i){
  j1 <- pairs[i,1]
  j2 <- pairs[i,2]
  mean(path$Th[new.list[[j1]],new.list[[j2]]])
})
Tt.small <- matrix(nrow = length(new.list), ncol = length(new.list))
Tt.small[rbind(pairs,pairs[,2:1])] <- tt.small
diag(Tt.small) <- 1
Tt.small #the smoothed kendall's tau matrix
```

Note that I compared the clusters grouped by industry and clusters grouped by this algorithm. Result is shown in figure 4. One could say that clusters formed by stocks from the same industry usually do not satisfy the PEA. In figure 4b, the circled 4 stocks would be used to represent their clusters and generate the results in section 4. Stock PGHN and ALC both have missing values, and thus are not inside any cluster.

(a) Clusters of industry

UBSG	LONN	ZURN	NOVN	SGSN
CSGN	GIVN	SREN	ROG	SCMN
LHN	SIKA	SLHN	UHR	
CFR	GEBN	ABBN	NESN	

(b) Clusters satisfying PEA

UBSG	LONN	ABBN	ZURN	ROG
CSGN	GIVN	SIKA	CFR	SCMN
NOVN	SREN	GEBN	UHR	PGHN
NESN	SLHN	LHN	SGSN	ALC

Figure 4

4 Attainability of Concordance Signatures

4.1 A quick look at the dependence

In this section, I investigate the dependence among the 13 clusters obtained in section 3.3 and the two stocks with missing values, PGHN and ALC. Instead of using the pairwise Kendall's tau correlation, I use a higher order measure of association, probabilities of concordance, and its vector form, concordance signature. Both of them are defined in McNeil et al. (2020).

As it would be introduced in section 4.2, given the partial concordance signature on a set $\mathcal{D}$, only some values of the remaining concordance signature is attainable, and this attainable set forms a convex hull. In this section, I study the attainable set of concordance signature on set $\mathcal{D}$ that includes PGHN, ALC, and some stocks from the 13 clusters, using the methodology introduced in paper McNeil et al. (2020). I explore the shape of this convex hull, study the impact when we add more stocks with no missing values into $\mathcal{D}$, check if the attainable set of concordance signature successfully captured the estimated concordance signature, and check if higher-order independence assumption is attainable in the case of this dataset.

To study the attainable set of the remaining concordance signature, I would first need to compute the partial concordance signature. McNeil et al. (2020) introduces two ways of doing so, one using the dataset directly, and one using the Kendall's correlation matrix. I will talk about both of them in section 4.3.

4.2 Atainable concordance signature

Definition 3 *Similar to Kendall's tau, the probabilities of concordance κ_I for the sub-vector $\mathbf{X}_I$ of vector $\mathbf{X}$, corresponding to subsets $I \subseteq \{1, ..., d\}$, is given by:*

$$\kappa_I = \kappa(X_I) = 2\mathbb{P}(\{\mathbf{X}_I < \mathbf{X}_I^*)\} \cup \{\mathbf{X}_I^* < \mathbf{X}_I\}),$$

where $\mathbf{X}_I^*$ denotes an independent copy of $\mathbf{X}$, $\{\mathbf{X}_I < \mathbf{X}_I^*\} = \{X_i < X_i^*, i \in I\}$.

The concordance signature of $\mathbf{X}$ is then defined to describe the vector $(\kappa_I : I \in \mathcal{P}(\mathcal{D}))$, where $\mathcal{P}(\mathcal{D})$ is the power set of $\mathcal{D} = \{1, ..., d\}$.

Note that pairwise concordance probabilities are related to Kendall's tau by $2\kappa_{\{i,j\}} - 1 = \tau_{\{i,j\}}$.

From McNeil et al. (2020), the probabilities of concordance of any subset $I \subseteq \mathcal{D}$ with odd cardinality can be deduced by

$$\kappa_I = 1 - \frac{|I|}{2} + \sum_{A \subset I, 2 \leq |A| < |I|} (-1)^{|A|} \frac{\kappa_A}{2}.$$

Thus, only probabilities of concordance of subsets with even cardinality is needed to determine the full concordance signature. The following theorem from McNeil et al. (2020) gives the set of attainable even concordance signatures (and so is the set of attainable full concordance signatures) of a d-dimensional vector.

Theorem 1 *The set of attainable even concordance signature of a d-dimensional vector is given by the convex hull*

$$\mathbb{K} = \left\{ \sum_{k=1}^{2^{d-1}} w_k \boldsymbol{\alpha}_k : w_k \geq 0, k = 1, ..., 2^{d-1}, \sum_{k=1}^{2^{d-1}} w_k = 1 \right\}, \tag{2}$$

where $\boldsymbol{\alpha}_k = (\alpha_{I,k} : I \in \mathcal{E}(\mathcal{D}))$ is given by

$$\alpha_{I,k} = \begin{cases} 1 & \text{if } I \subseteq J_k \text{ or } I \subseteq J_k^C \\ 0 & \text{otherwise} \end{cases}.$$

$\mathcal{E}(\mathcal{D})$ is the subsets of even cardinality, J_k is a set of indices enumerated by $k \in \{1, ..., 2^{d-1}\}$, corresponding to the 2^{d-1} different combinations of J_k.

Note that the vectors $\boldsymbol{\alpha}_k$, $k \in \{1, ..., 2^{d-1}\}$ are linearly independent by (McNeil et al., 2020, Proposition 5(ii)).

To prove this theorem, we would need to introduce extremal copulas and extremal mixture copulas, as well as their two propositions. The definitions and proofs could be found in McNeil et al. (2020):

Proposition 2 *1. For any $k \in \{1, ..., 2^{d-1}\}$ and $I \in \mathcal{P}(\mathcal{D})$, the probabilities of concordance of the kth extremal copula is given by $\kappa(C_I^{(k)}) = \alpha_{I,k}$.*

2. For an extremal mixture copula $C = \sum_{k=1}^{2^{d-1}} w_k C^{(k)}$, the probabilities of concordance for any $I \subseteq \mathcal{D}$ is

$$\kappa(C_I) = \sum_{k=1}^{2^{d-1}} w_k \kappa(C_I^{(k)}) = \sum_{k=1}^{2^{d-1}} w_k \alpha_{I,k}.$$

Proposition 3 *$\boldsymbol{\kappa}(C) = \sum_{k=1}^{2^{d-1}} w_k \boldsymbol{\alpha}_k$ uniquely determines C.*

These two propositions show that equation 2 is the set of attainable even corcondance signatures of d-dimensional extremal mixture copulas. And these sets are also the sets of attainable even concordance signatures for any d-dimensional copula, because by McNeil et al. (2020):

Lemma 1 *For a d-dimensional copula C with even concordance signature $\boldsymbol{\kappa}(C)$, there exists a unique etremal mixture copula $C^* = \sum_{k=1}^{2^{d-1}} w_k C^{(k)}$ such that $\boldsymbol{\kappa}(C) = \boldsymbol{\kappa}(C^*)$. The weights w_k are given by the unique solution of the linear equation system*

$$\boldsymbol{\kappa}(C) = \sum_{k=1}^{2^{d-1}} w_k \boldsymbol{\alpha}_k.$$

An application to theorem 1 is to check if a given partial concordance signature is attainable, and if it is attainable, determine the sets of attainable remaining concordance signature. For example, in the case of SMI, stock PGHN and ALC have missing values. This would prevent us from calculating the probabilities of concordance on the subsets including these two stocks. To solve this problem, one can do the following:

First introduce a $2^{d-1} \times 2^{d-1}$ matrix A_d with $\boldsymbol{\alpha}_k$ as its columns, where d is the dimension of the dataset. In this case, d would be 15 (13 clusters obtained in section 3 plus 2 stocks PGHN and ALC). Then, let $A_d^{(1)}$ be the $|S| \times 2^{d-1}$ matrix, consisting of the rows of A_d that correspond to S, where $S \subset \mathcal{E}(\mathcal{D})$ represents the subsets whose probabilities of concordance are already obtained. Thus, the set of weight vectors $\{\boldsymbol{w} : A_d^{(1)} \boldsymbol{w} = \boldsymbol{\kappa}, \boldsymbol{w} \geq 0\}$, where $\boldsymbol{\kappa} = (\kappa_I : I \in S)$ denotes the partial concordance signature we already have, should satisfy the sum condition on the weights, since S is a label set containing $\emptyset$ and the first row of $A_d^{(1)}$ consists of ones. The set that satisfies theses conditions should be of the form:

$$\left\{ \sum_{i=1}^{m} \mu_i \boldsymbol{w}_i : \sum_{i=1}^{m} \mu_i = 1, \mu_i \geq 0, i = 1, ..., m \right\}. \tag{3}$$

The vertices of this convex polytope, $\boldsymbol{w}_1, ..., \boldsymbol{w}_m$, could be found using the method of Avis and Fukuda (1992). And the set of attainable unsepcified elements of the even concordance signatures is given by the convex hull

$$\{\boldsymbol{\kappa}_1^{(2)} = A_d^{(2)} \boldsymbol{w}_i, i = 1, ..., m\}, \tag{4}$$

where $A_d^{(2)}$ is the matrix formed of the remaining rows of A_d.

4.3 Estimating concordance signature

McNeil et al. (2020) provides two ways to obtain the concordance signatures.

Firstly, the concordance signature is identical for the copulas of all elliptical distributions with the same correlation matrix P, which is twice the orthant probabilities for a centred elliptical distribution with dispersion matrix P. Thus, one could generate its concordance signature by calculating the orthant probabilities of a multivariate normal distribution $\mathbf{X} \sim N_d(\mathbf{0}, P)$.

Since the residuals of the 18 stocks follow an ellipitcal distribution, the partial concordance signatures on the subsets of the 13 clusters could be generated directly from the corresponding smoothed kendall's tau matrix we obtained in section 3.

Secondly, for a random sample $\mathbf{X}_1, ..., \mathbf{X}_n$ with continuous margins and no ties, McNeil et al. (2020) also gives a way to estimate the concordance signature $\boldsymbol{\kappa}$:

$$\kappa_{I,n} = \frac{2}{n(n-1)} \sum_{i \neq j} \prod_{k \in I} I_{\{X_{i,k} \leq X_{j,k}\}}. \tag{5}$$

This is obtained by plugging the estimater of the multivariate Kendall's tau coefficients $\tau(C_I)$ into

$$(2^{|I|-1} - 1)\tau(C_I) = 2^{|I|-1}\kappa(C_I) - 1,$$

given by Genest et al. (2011)

This would allow us to estimate the concordance signature using the dataset as long as every marginal vectors is of the same length. Stock PGHN has 70 instances, after the date September 18th. Using these last 70 instances, one could estimate the concordance signature among PGHN and other stocks.

4.4 Main Results

In total, there are 13 clusters and 2 stocks with missing values. This would make the dataset to be dimension 15. This dataset is still too large, and dealing with all of the 15 vectors would require a very long running time. So in this project, I only deal with a small part of the dataset. The conclusions could be generalized to a higher dimension.

We are interested in the concordance signature on the set $\mathcal{D}$ which contains part of the clusters and the stocks PGHN and ALC. I mainly focus on $\mathcal{D}$ containing only PGHN, and the results are the same for $\mathcal{D}$ containing only ALC. I explore a little bit on $\mathcal{D}$ with both PGHN and ALC in the end, but the result would not be very constructive since we only study the concordance signature on a small $\mathcal{D}$ and in low dimension projections.

Note that since any stock could represent its entire cluster, I refer to each cluster by a component inside it (the circled four stocks in figure 4b). Below shows the sample code for calculating the concordance signature on set $\mathcal{D} = \{\text{UBSG}, \text{LHN}, \text{LONN}\}$ as well as generating the attainable even concordance signature when we add stock PGHN into $\mathcal{D}$. More details could be found in https://github.com/ajmcneil/KendallSignature.

```
library(KendallSignature)
library(copula)
library(rcdd)
library(gMOIP)
library(rgl)

Y = Tt.small[1:3,1:3] # kendalls matrix of UBSG, LHN, LONN
# calculating concordance signature
k = consignature(Y)
# find vertices of the set of weight vectors
w = findpolytope(k, d = 4)
# get the vertices of the attainable even concordance signature of I
allkappa = Amatrix(4) %*% t(w)
# select the vertices for missing concordance probabilities
vertices = t(allkappa[c(4,6:8), ])

#plot attainable set
plot3d(vertices[,1:3],xlab='PGHN-UBSG',ylab='PGHN-LHN',zlab='PGHN-LONN')
plotHull3D(clean.dupli(vertices[,1:3]),drawPoints=TRUE)
plotHull2D(clean.dupli(vertices[,1:2]),drawPoints=TRUE)
          +xlab('PGHN-UBSG')+ylab('PGHN-LHN')
```

```
#clean.dupli() is a function I wrote to clean up duplicated
   vertices. plotHull functions could not handle duplicated values.
```

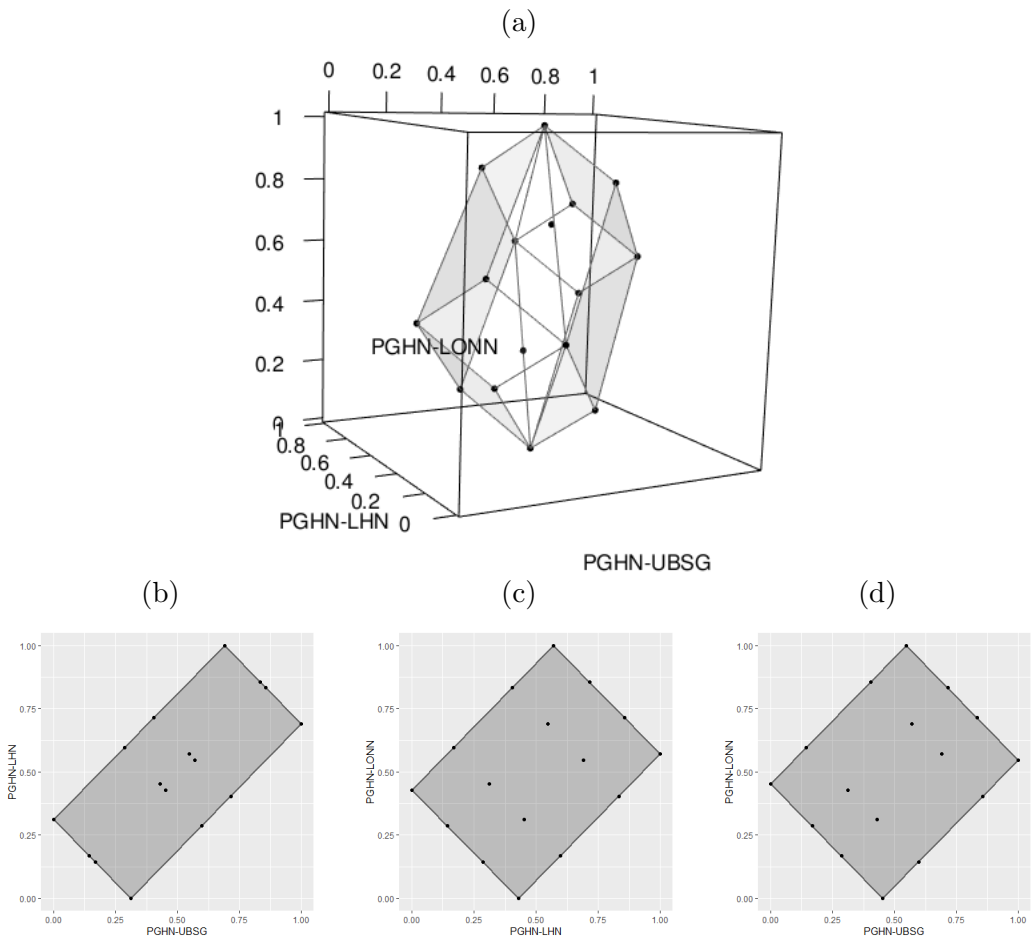


Figure 5: (a) is the convex hull of the set of attainable pairwise concordance on set $\mathcal{D}$, (b)(c)(d) are its 2D projections

The code above would return the 16 vertices of the convex hull of the set of attainable even concordance signature on $\mathcal{D}$. The set of the attainable pairwise concordance signatures, as well as its 2D projections, are plotted in figure 5. One could notice from these 4 plots that some 'vertices' are inside the boundary of the convex hull of the attainable sets, and thus no longer remain as vertices. Indeed, as we add more stocks into $\mathcal{D}$, the convex hull of the sets of attainable concordance signature would become more complicated, but its lower dimension projections would remain the same.

This could be shown by a 2D example on the subset $I = \{\text{PGHN}, \text{LHN}, \text{UBSG}\}$, while we add more stocks into the set $\mathcal{D}$. In figure 6, I plotted the sets of attainable pairwise probabilities of concordance between PGHN and UBSG, against the sets of attainable pairwise probabilities of concordance between PGHN and LHN. The attainable sets are computed based on set $\mathcal{D} = \{\text{PGHN}, \text{UBSG}, \text{LHN}\}$, $\mathcal{D} = \{\text{PGHN}, \text{UBSG}, \text{LHN}, \text{LONN}\}$, and $\mathcal{D} = \{\text{PGHN}, \text{UBSG}, \text{LHN}, \text{LONN}, \text{SIKA}\}$ correspondingly, and results are plotted in figure 6a, 6b, and 6c. As the size of $\mathcal{D}$ increases, more 'vertices' appear inside the boundaries of the 2D convex hull. They are all vertices of the convex hull of the set of attainable concordance signature in a higher dimension, but no longer remain as one in its 2D projections. Thus, one could conclude that adding more stocks to set $\mathcal{D}$ as a restriction would affect the entire set of attainable concordance signature, but will not affect the partial set of attainable concordance signature corresponding to the old set.

(a) (b) (c)

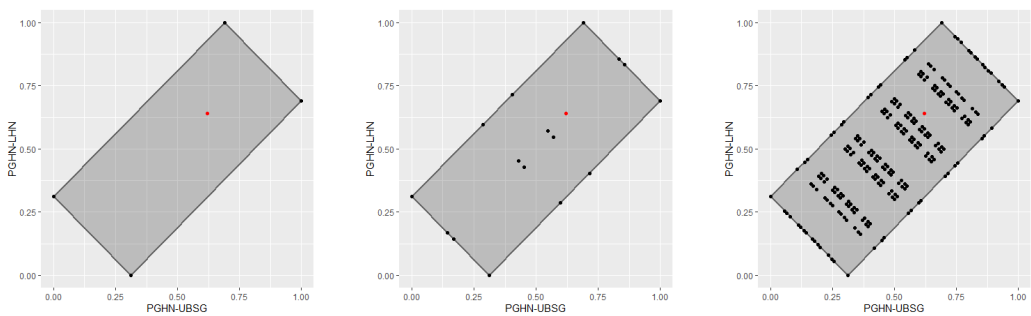


Figure 6: The 2D projections of the set of attainable pairwise concordance signature among PGHN, LHN, UBSG, when d=3(a), d=4(b), and d=5(c)

Also, I estimated the pairwise concordance signature among PGHN, LHN, and UBSG, using 70 data instances begining September 18th. The result is printed below and plotted as a red dot in the corresponding plots in figure 6.

```
# X is the matrix of residuals
> Y = cbind(X[182:251,c(1,3)],gfit_PGHN@fit$z)
> estsignature(as.matrix(Y))

    empty     {1,2}     {1,3}     {2,3}
1.0000000 0.7461698 0.6202899 0.6397516
```

I stop at $|\mathcal{D}| = 5$, because for a $\mathcal{D}$ with larger size, it would take too long running time and would result in more than 30000 vertices. This is because

as $|\mathcal{D}|$ increases, the number of even subsets would increase exponentially. However, if we could set $\kappa_I = 0$ for all I with cardinality larger than 2, we could weaken this effect and it would allow us to deal with set $\mathcal{D}$ of larger size. Indeed, it is common to assume higher-order independence, and I would test if this assumption is possible on our dataset.

On set $\mathcal{D} = \{$UBSG, LHN, LONN, SIKA, ABBN $\}$, I first estimated the even concordance signature using their data, then set $\kappa_I = 0$ for every $|I| = 4$, and tested its attainability using the following code. Unfortunately, this assumption is not attainable in this case, and not attainable using $\mathcal{D}$ with different sizes and different components.

```
> Y = X[c(1,3,4,6,7)]
> kappa = estsignature(as.matrix(Y))
> kappa[12:16]=0
> attainable(kappa)

Error during wrapup: Solution on simplex not found
```

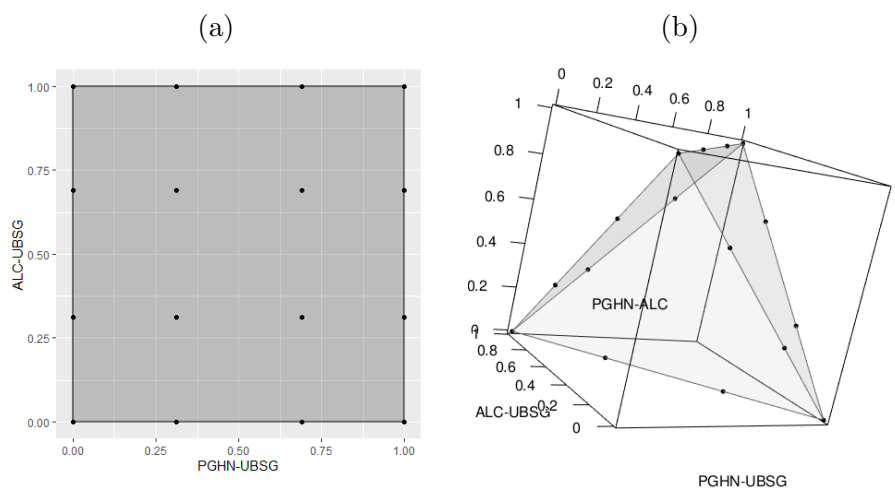


Figure 7: Attainable concordance signature on set $\mathcal{D} = \{\text{PGHN}, \text{ALC}, \text{UBSG}, \text{LHN}\}$

In the end, I also studied the concordance signature on set $\mathcal{D}$ that includes both stocks with missing values, PGHN and ALC. Figure 7 shows the 2D and 3D projection of the set of attainable pairwise concordance signature among stocks ALC, PGHN and UBSG. One could notice that compared to the previous results, this set of attainable concordance signatures has a wider range.

And from the 2D projection, one could discover that every point in $[0,1]^2$ is attainable.

A more constructive result would require a look at a higher dimension convex hull on a $\mathcal{D}$ with larger size, which is complicated to compute and visualize, so I leave this for further research.

5 Conclusion

In this project, I preprocessed the data, reduced data dimension, and focused on the set of attainable concordance signature among the stocks with missing values and stocks without missing values. I discovered that the convex hull of the set of attainable concordance signature on a particular set would not change as we add more stocks into account, and that the common assumption of higher order independence is usually not attainable. The biggest problem I faced is the running time. Since running time could increase exponentially as the dimension of data increases, I could only deal with a very small part of the entire dataset, which prevents me from gaining a further understanding when there is more than one stock with missing values.

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